

Regression Models

Applied Machine Learning — Session 2, Chapter 1

Regression: What We're Doing

Predicting a continuous number.

Examples:

- House price given area, location, rooms
- Energy consumption given temperature, day of week
- Patient recovery time given treatment, age, biomarkers

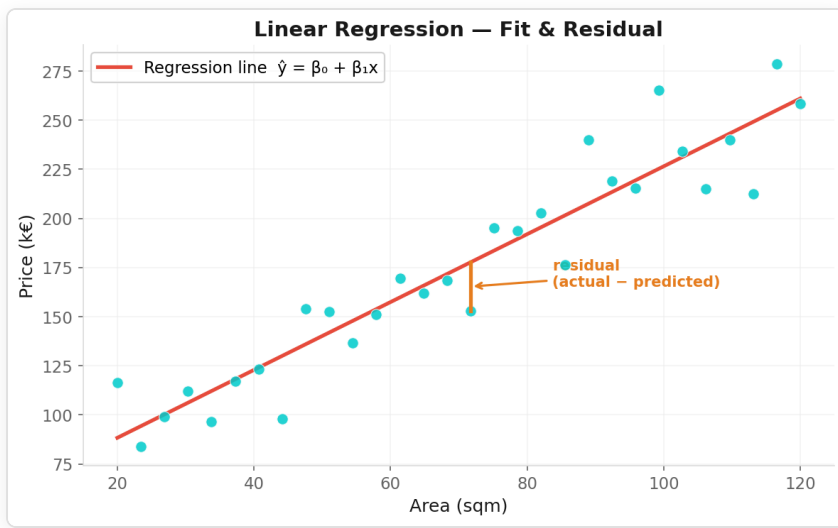
How we measure success:

- MAE: average absolute error
- RMSE: penalizes large errors
- R^2 : how much variance we explain

Linear Regression

Model: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$

Learning: Minimize sum of squared residuals (OLS)



Linear Regression

Model: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)

print(model.coef_)      # feature weights
print(model.intercept_) # bias
```

Strengths: Fast, interpretable, great baseline

Weakness: Assumes linear relationship

Interpreting Coefficients

```
for feature, coef in zip(feature_names, model.coef_):  
    print(f'{feature}: {coef:.2f}')
```

Reading: If `area_sqm` has coefficient 3.5:

→ Each additional square meter → +€3,500 in predicted price

(all other features held constant)

This is the power of linear models: **interpretability**.

Polynomial Regression

What if the relationship is curved?

Add polynomial features: x , x^2 , x^3 , ...

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import Pipeline

model = Pipeline([
    ('poly', PolynomialFeatures(degree=2)),
    ('linear', LinearRegression())
])
```

⚠ Higher degree → risk of overfitting

→ Always cross-validate!

Regularization: Why?

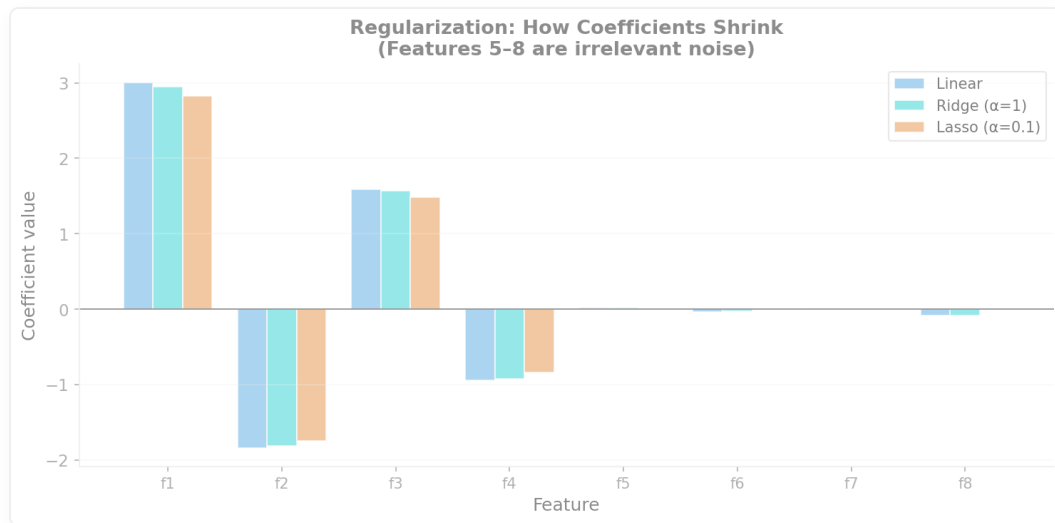
With many features, linear models can overfit.

Idea: Penalize large coefficients.

	Ridge (L2)	Lasso (L1)
Penalty	$\alpha \times \sum \beta_i^2$	$\alpha \times \sum \beta_i $
Effect	Shrinks β toward 0	Sets some β to exactly 0
Use when	All features useful	Many irrelevant features

```
from sklearn.linear_model import Ridge, Lasso
ridge = Ridge(alpha=1.0) # alpha = regularization strength
lasso = Lasso(alpha=0.1)
```

Regularization Intuition



With Ridge (L2): Keeps all features but smaller weights.

With Lasso (L1): Removes irrelevant features entirely → automatic feature selection.

α (alpha): Higher = stronger regularization = simpler model.

Decision Tree Regression

Idea: Split the feature space into regions, predict the mean in each region.

```
Is area_sqm > 100?  
  YES → Is rooms > 3? → predict 420k / predict 280k  
  NO  → predict 195k
```

```
from sklearn.tree import DecisionTreeRegressor  
dt = DecisionTreeRegressor(max_depth=3)
```

- No feature scaling needed
- Captures non-linearity and interactions
- ⚠ Prone to overfitting → limit `max_depth`

Random Forest

Wisdom of crowds — for models.

- Train many decision trees on random subsets of data + features
- Predict = average of all trees

```
from sklearn.ensemble import RandomForestRegressor
rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

# Bonus: feature importance!
importances = rf.feature_importances_
```

Almost always better than a single tree. 

Choosing a Regression Model

Model	Interpretable	Non-linear	Scaling needed	Handles outliers
Linear Regression	✓	✗	✓	✗
Ridge / Lasso	✓	✗	✓	⚠
Decision Tree	✓	✓	✗	✓
Random Forest	⚠	✓	✗	✓

Rule of thumb: Start with Linear/Ridge as baseline. Add tree models if non-linearity suspected.

Now: Exercises!

→ Open `03-exercises/ch04_regression_exercises.ipynb`

Task: Predict diabetes disease progression using four regression models.
Compare their performance. Which model wins?

~10 minutes

Key Takeaways

- Linear regression: interpretable, powerful baseline
- Polynomial regression: handles curves, watch for overfitting
- Ridge/Lasso: regularization prevents overfitting
- Random Forest: robust, non-linear, feature importances
- Always compare multiple models with cross-validation

Next: Chapter 5

Classification Models

"What if the answer is a category, not a number?"